

**THE UNIVERSITY OF AZAD JAMMU & KASHMIR,  
MUZAFFARABAD**

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**Roll No**

**2024-SE-24**

**Semester**

**04**

**Course Title**

**DBMS**

**Submitted By**

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**Lab**

**02**

# Database: Point of Sale System

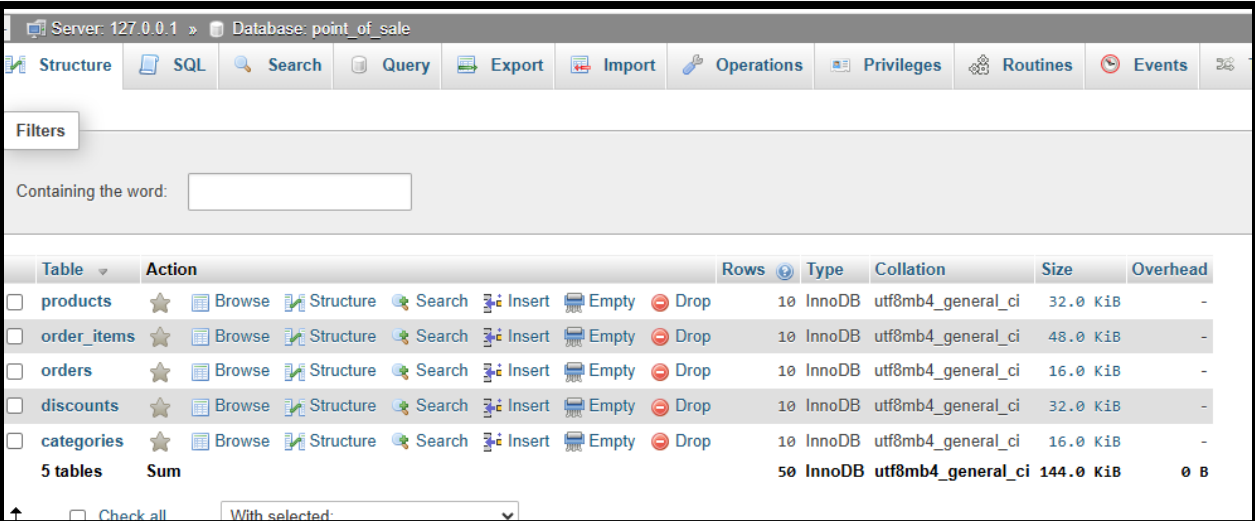
## 1. Introduction

In this lab, we designed a Point of Sale (POS) database system. The database simulates a small retail environment where products are categorized, customers place orders, and discounts are applied. The goal is to practice creating tables, defining relationships, inserting sample data, and running queries to extract useful information.

## 2. Database Structure

The database consists of the following tables:

- **Roles** – Defines user roles (Admin, Customer, Salesman).
- **Categories** – Groups products into categories such as Beverages, Snacks, Dairy, etc.
- **Products** – Stores product details including name, price, and category.
- **Orders** – Records customer orders with date and customer name.
- **Order Items** – Links products to orders with quantities.
- **Discounts** – Stores discount percentages for products.



| Table                                | Action                                      | Rows | Type   | Collation          | Size      | Overhead |
|--------------------------------------|---------------------------------------------|------|--------|--------------------|-----------|----------|
| <input type="checkbox"/> products    | ★ Browse Structure Search Insert Empty Drop | 10   | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> order_items | ★ Browse Structure Search Insert Empty Drop | 10   | InnoDB | utf8mb4_general_ci | 48.0 KiB  | -        |
| <input type="checkbox"/> orders      | ★ Browse Structure Search Insert Empty Drop | 10   | InnoDB | utf8mb4_general_ci | 16.0 KiB  | -        |
| <input type="checkbox"/> discounts   | ★ Browse Structure Search Insert Empty Drop | 10   | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> categories  | ★ Browse Structure Search Insert Empty Drop | 10   | InnoDB | utf8mb4_general_ci | 16.0 KiB  | -        |
| 5 tables                             | Sum                                         | 50   | InnoDB | utf8mb4_general_ci | 144.0 KiB | 0 B      |

## 3. Data Insertion

Sample data was inserted into each table:

- **Categories:** 10 categories (Beverages, Snacks, Dairy, Bakery, Fruits, Vegetables, Frozen, Meat, Seafood, Grains).
- **Products:** 10 products (Coke, Pepsi, Chips, Milk, Bread, Apple, Carrot, Ice Cream, Chicken, Fish).

SELECT \* FROM `products`

☐ Profiling

[\[ Edit inline \]](#)

[\[ Edit \]](#)

[\[ Explain SQL \]](#)

[\[ Create PHP code \]](#)

[\[ Refresh \]](#)

☐ Show all

Number of rows: 25

Filter rows:

Extra options

<

## Orders and Order Items

- **Orders:** 10 orders placed by customers (Ali, Ahmed, Sara, Zain, Ayesha, Usman, Hina, Bilal, Fatima, Omar).
- **Order Items:** Each order linked with products and quantities.

Server: 127.0.0.1 » Database: point\_of\_sale » Table: orders

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key:

Extra options

|                          |      |      |        | order_id | order_date | customer_name |
|--------------------------|------|------|--------|----------|------------|---------------|
| <input type="checkbox"/> | Edit | Copy | Delete | 1        | 2026-01-01 | Ali           |
| <input type="checkbox"/> | Edit | Copy | Delete | 2        | 2026-01-02 | Ahmed         |
| <input type="checkbox"/> | Edit | Copy | Delete | 3        | 2026-01-03 | Sara          |
| <input type="checkbox"/> | Edit | Copy | Delete | 4        | 2026-01-04 | Zain          |
| <input type="checkbox"/> | Edit | Copy | Delete | 5        | 2026-01-05 | Ayesha        |
| <input type="checkbox"/> | Edit | Copy | Delete | 6        | 2026-01-06 | Usman         |
| <input type="checkbox"/> | Edit | Copy | Delete | 7        | 2026-01-07 | Hina          |
| <input type="checkbox"/> | Edit | Copy | Delete | 8        | 2026-01-08 | Bilal         |
| <input type="checkbox"/> | Edit | Copy | Delete | 9        | 2026-01-09 | Fatima        |
| <input type="checkbox"/> | Edit | Copy | Delete | 10       | 2026-01-10 | Omar          |

☐ Check all | With selected: Edit Copy Delete Export

- **Discounts:** Discounts are applied to products. Each products can have a discount percentage.

Server: 127.0.0.1 » Database: point\_of\_sale » Table: discounts

Browse

Structure

SQL

Search

Insert

Export

In

Showing rows 0 - 9 (10 total, Query took 0.0004 seconds.)

SELECT \* FROM `discounts`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

Extra options

discount\_id

product\_id

discount\_percent

Edit

Copy

Delete

1

1

10

Edit

Copy

Delete

2

2

5

Edit

Copy

Delete

3

3

15

Edit

Copy

Delete

4

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8

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12

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Delete

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8

20

Edit

Copy

Delete

9

9

10

Edit

Copy

Delete

10

10

5

Check all

With selected:

Edit

Copy

Delete

Export

Table structure | Relation view

| #                          | Name       | Type    | Collation | Attributes | Null | Default | Comments | Extra          | Action             |
|----------------------------|------------|---------|-----------|------------|------|---------|----------|----------------|--------------------|
| <input type="checkbox"/> 1 | item_id    | int(11) |           |            | No   | None    |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> 2 | order_id   | int(11) |           |            | Yes  | NULL    |          |                | Change  Drop  More |
| <input type="checkbox"/> 3 | product_id | int(11) |           |            | Yes  | NULL    |          |                | Change  Drop  More |
| <input type="checkbox"/> 4 | quantity   | int(11) |           |            | Yes  | NULL    |          |                | Change  Drop  More |

☐ Check all    With selected: Browse Change Drop Primary Unique Index Spatial Fulltext  
 Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize  
 Add 1 column(s) after quantity Go

Indexes

| Action             | Keyname    | Type  | Unique | Packed | Column     | Cardinality | Collation | Null | Comment |
|--------------------|------------|-------|--------|--------|------------|-------------|-----------|------|---------|
| Edit  Rename  Drop | PRIMARY    | BTREE | Yes    | No     | item_id    | 10          | A         | No   |         |
| Edit  Rename  Drop | order_id   | BTREE | No     | No     | order_id   | 10          | A         | Yes  |         |
| Edit  Rename  Drop | product_id | BTREE | No     | No     | product_id | 10          | A         | Yes  |         |

Create an index on 1 columns Go

Activate Windows  
Go to Settings to activate Windows

Table structure | Relation view

| #                          | Name        | Type          | Collation          | Attributes | Null | Default | Comments | Extra          | Action             |
|----------------------------|-------------|---------------|--------------------|------------|------|---------|----------|----------------|--------------------|
| <input type="checkbox"/> 1 | product_id  | int(11)       |                    |            | No   | None    |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> 2 | name        | varchar(100)  | utf8mb4_general_ci |            | Yes  | NULL    |          |                | Change  Drop  More |
| <input type="checkbox"/> 3 | price       | decimal(10,2) |                    |            | Yes  | NULL    |          |                | Change  Drop  More |
| <input type="checkbox"/> 4 | category_id | int(11)       |                    |            | Yes  | NULL    |          |                | Change  Drop  More |

☐ Check all    With selected: Browse Change Drop Primary Unique Index Spatial Fulltext  
 Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize  
 Add 1 column(s) after category\_id Go

Indexes

| Action             | Keyname     | Type  | Unique | Packed | Column      | Cardinality | Collation | Null | Comment |
|--------------------|-------------|-------|--------|--------|-------------|-------------|-----------|------|---------|
| Edit  Rename  Drop | PRIMARY     | BTREE | Yes    | No     | product_id  | 10          | A         | No   |         |
| Edit  Rename  Drop | category_id | BTREE | No     | No     | category_id | 10          | A         | Yes  |         |

## 4. Queries and Results

- **Total Orders**

This query shows the total number of orders placed.

Your SQL query has been executed successfully.

```
-- Total orders SELECT COUNT(*) AS Total_Orders FROM Orders;
```

- **Total Sales per Product**

This query calculates how many units of each product were sold.

✓ Showing rows 0 - 9 (10 total, Query took 0.0007 seconds.)

```
-- Total sales per product SELECT p.name, SUM(oi.quantity) AS total_sold FROM Order_Items oi JOIN Products p ON oi.product_id = p.product_id GROUP  
BY p.name;
```

[ [Edit inline](#) ] [ [Edit](#) ] [ [Create PHP code](#) ]

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Extra options

| name      | total_sold |
|-----------|------------|
| Apple     | 3          |
| Bread     | 1          |
| Carrot    | 4          |
| Chicken   | 2          |
| Chips     | 5          |
| Coke      | 2          |
| Fish      | 1          |
| Ice Cream | 1          |
| Milk      | 2          |
| Pepsi     | 1          |

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Activate Windows

- **Orders with Customer Names**

Displays all orders along with customer names and dates.

✓ Showing rows 0 - 9 (10 total, Query took 0.0004 seconds.)

-- Orders with customer names `SELECT order_id, customer_name, order_date FROM Orders;`

[ Edit inline ] [ Edit ] [ Create PHP code ]

☐ Show all

Number of rows: 25

Filter rows:

Sort by key: None

Extra options

← T →

order\_idcustomer\_nameorder\_date

|                          |                  |          |            |
|--------------------------|------------------|----------|------------|
| <input type="checkbox"/> | Edit Copy Delete | 1 Ali    | 2026-01-01 |
| <input type="checkbox"/> | Edit Copy Delete | 2 Ahmed  | 2026-01-02 |
| <input type="checkbox"/> | Edit Copy Delete | 3 Sara   | 2026-01-03 |
| <input type="checkbox"/> | Edit Copy Delete | 4 Zain   | 2026-01-04 |
| <input type="checkbox"/> | Edit Copy Delete | 5 Ayesha | 2026-01-05 |
| <input type="checkbox"/> | Edit Copy Delete | 6 Usman  | 2026-01-06 |
| <input type="checkbox"/> | Edit Copy Delete | 7 Hina   | 2026-01-07 |
| <input type="checkbox"/> | Edit Copy Delete | 8 Bilal  | 2026-01-08 |
| <input type="checkbox"/> | Edit Copy Delete | 9 Fatima | 2026-01-09 |
| <input type="checkbox"/> | Edit Copy Delete | 10 Omar  | 2026-01-10 |

↑

☐ Check all

With selected: Edit Copy Delete Export

Activate Windows  
Go to Settings to activate Windows

- **Products with Discounts**

Shows products with their respective discount percentages.



The screenshot shows a database query result interface. At the top, a green status bar indicates 'Showing rows 0 - 9 (10 total, Query took 0.0006 seconds.)'. Below this, the SQL query is displayed: `-- Products with discounts SELECT p.name, d.discount_percent FROM Products p JOIN Discounts d ON p.product_id = d.product_id;`. There are links for '[ Edit inline ]', '[ Edit ]', and '[ Create PHP code ]'. Below the query, there are controls for 'Show all' (checkbox), 'Number of rows' (set to 25), 'Filter rows' (a search box with 'Search this table'), and 'Sort by key' (set to None). An 'Extra options' button is also present. The main table has two columns: 'name' and 'discount\_percent'. It lists 10 products with their respective discount percentages.

| name      | discount_percent |
|-----------|------------------|
| Coke      | 10               |
| Pepsi     | 5                |
| Chips     | 15               |
| Milk      | 8                |
| Bread     | 12               |
| Apple     | 7                |
| Carrot    | 6                |
| Ice Cream | 20               |
| Chicken   | 10               |
| Fish      | 5                |

These queries demonstrate how the POS database can provide insights into sales and customer activity.

## 5. ER Diagram Explanation

The Entity-Relationship (ER) Diagram visually represents the structure of the database:

- **Roles:** Independent table, not directly linked to others in this lab.
- **Categories → Products:** One category can have many products (1-to-many relationship).
- **Products → Order\_Items:** A product can appear in many order items (1-to-many).
- **Orders → Order\_Items:** One order can contain multiple items (1-to-many).
- **Products → Discounts:** Each product can have one discount (1-to-1 relationship in this design).

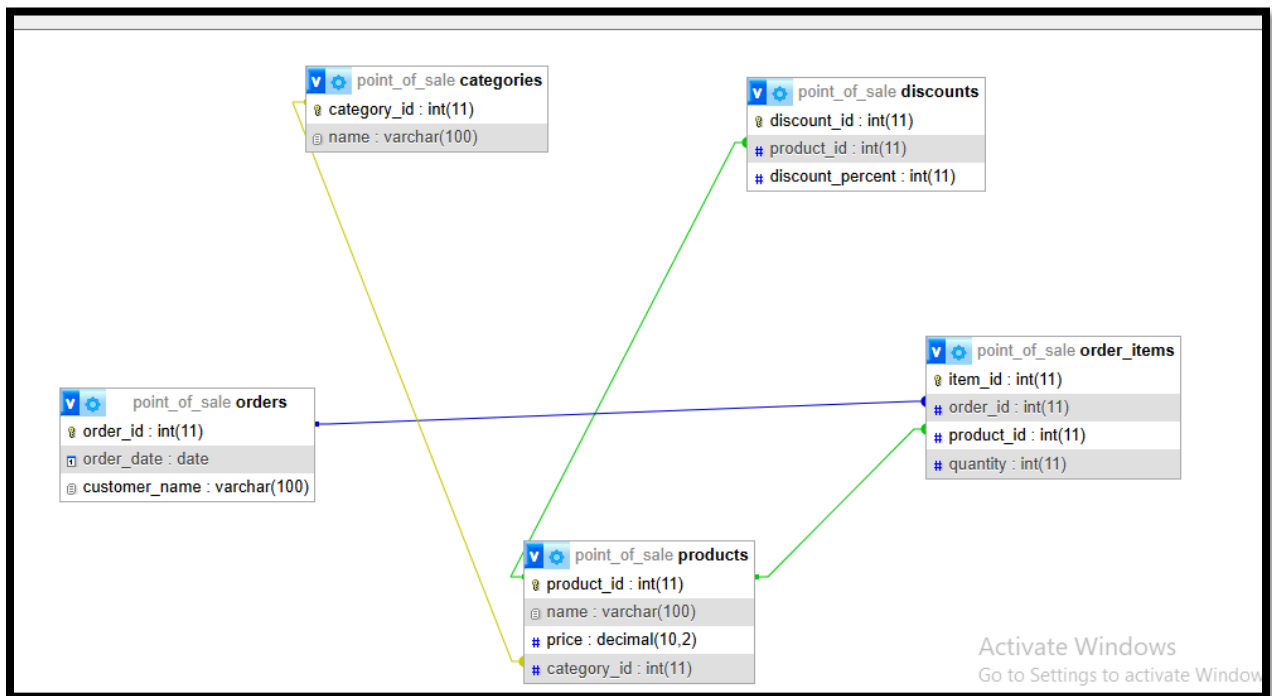
### ➤ ER Diagram (Conceptual Description)

**Entities:** Roles, Categories, Products, Orders, Order\_Items, Discounts.

**Relationships:**

- **Categories** → Products (1:N)
- **Orders** → Order\_Items (1:N)
- **Products** → Order\_Items (1:N)
- **Products** → Discounts (1:1)

This diagram shows how data flows between customers, orders, products, and discounts in the POS system.



Exporting tables from "point\_of\_sale" database

**Export templates:**

**New template:**

Template name:

**Existing templates:**

Template:

**Export method:**

☒ Quick - display only the minimal options  
☐ Custom - display all possible options

**Format:**

Activate Windows  
Go to Settings to activate Windows.

## 6. Conclusion

This lab successfully demonstrates how to build a Point of Sale database with proper relationships between tables. The queries provide meaningful business insights such as total orders, product sales, customer details, and discounts. The ER diagram further clarifies the logical structure of the system, making it easier to understand and extend in future projects.

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